

Signal Processing

Exercise Sheet 11– FIR Filter

SoSe 2026

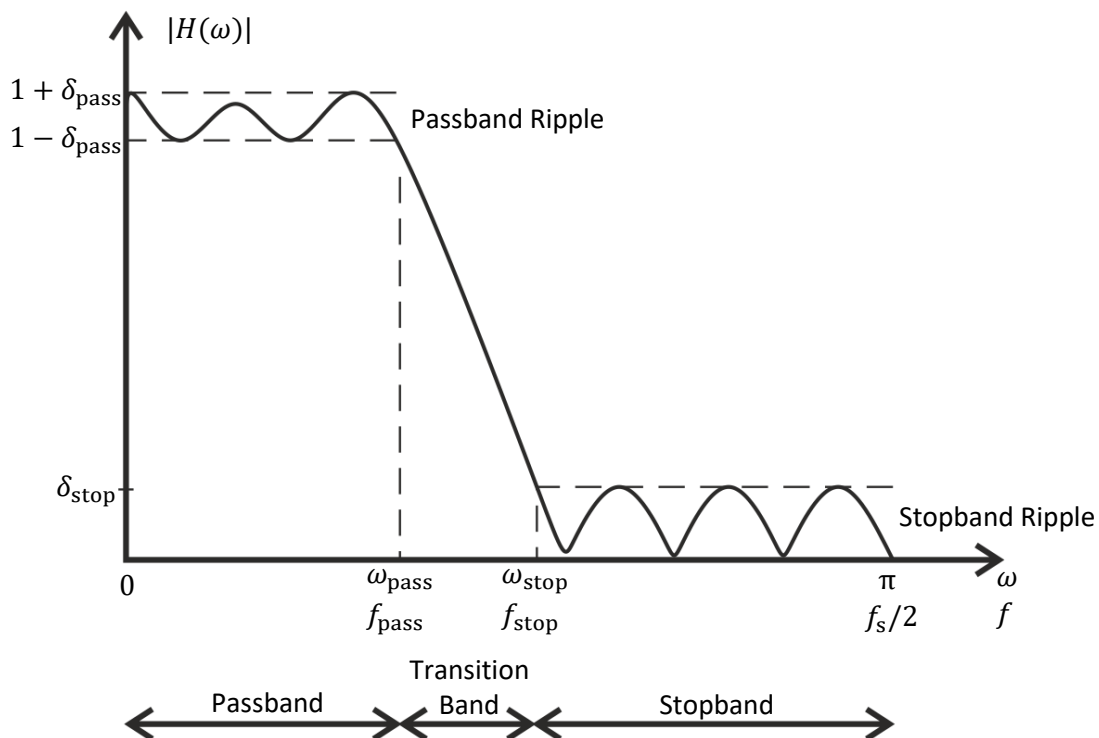
Introduction:

Why Do We Need Filters?

Most of the time, the final goal of using a filter is to achieve some kind of frequency selectivity on the spectrum of the input signal.

Examples: Elimination of 50 Hz noise (Notch filter), Anti-Aliasing filter (low-pass filter), elimination of DC Component (high-pass filter), ...

The figure shows an example of a practical low-pass filter. In this example, frequency components in the passband, from DC to ω_{pass} , will pass the filter with almost no attenuation. The components in the stopband, above ω_{stop} , will be suppressed significantly. Note that the frequency response of a practical filter cannot be absolutely flat in the passband or in the stopband. As shown in the figure, some ripples will be unavoidable and the transition band, $\omega_{\text{pass}} < \omega < \omega_{\text{stop}}$, cannot be infinitely sharp in practice.



Digital Filter Design in Four Steps

1) Determine specifications

Define the required specifications depending on the application.

2) Find a system function

The rational system function of a digital filter is:

$$H(z) = \frac{\sum b_k z^{-k}}{\sum a_k z^{-k}}.$$

This step calculates the coefficients a_k and b_k .

3) Choose a realization structure

Now that $H(z)$ is known, choose a realization structure, e.g., direct form I, II, cascade, parallel, transposed, lattice. Problem: Quantization errors.

4) Implement the filter

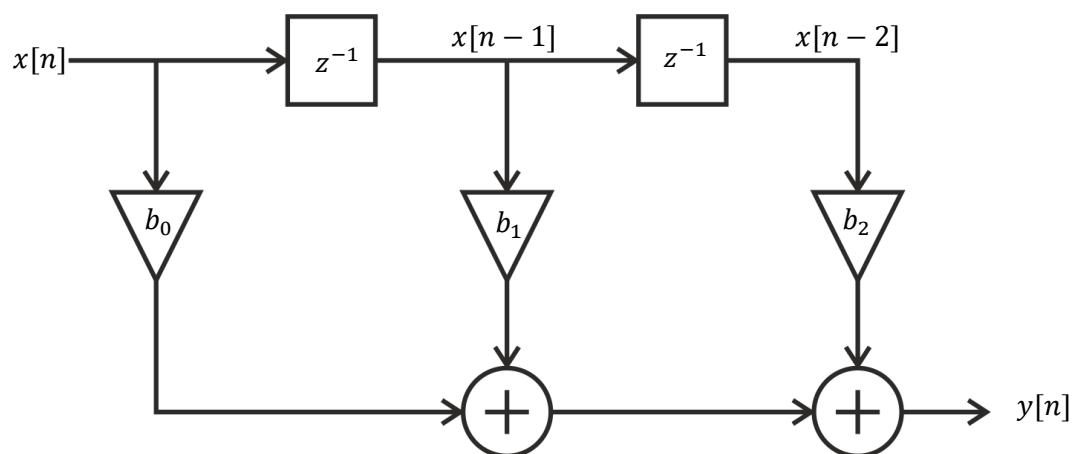
Software or Hardware implementation.

FIR Filters

An FIR filter is a special case of the before mentioned equation, where $a_0 = 1$ and $a_k = 0$ for $k = 1, \dots, N$, hence we obtain:

$$H(z) = \sum b_k z^{-k}.$$

The direct form realization of the equation is shown in the following figure.



As shown in the figure, a digital filter can be implemented using only three elements: Addition, Multiplication by a constant and delay blocks.

There are three coefficients and two delay blocks in the figure: The filter is of order 2 (number of delay blocks) and has a length of 3 (number of coefficients).

An FIR filter has two important advantages over an IIR design:

Firstly, there is no feedback loop in the structure of an FIR filter. Therefore, FIR filters are inherently stable. Meanwhile, for an IIR filter, we may need to check the stability.

Secondly, an FIR filter can provide a linear-phase response. As a matter of fact, a linear-phase response is the main advantage of an FIR filter over an IIR design – otherwise, for the same filter specifications, an IIR filter will lead to a lower order. In order to have a linear-phase FIR filter, we must provide symmetry in the time domain of the system function.

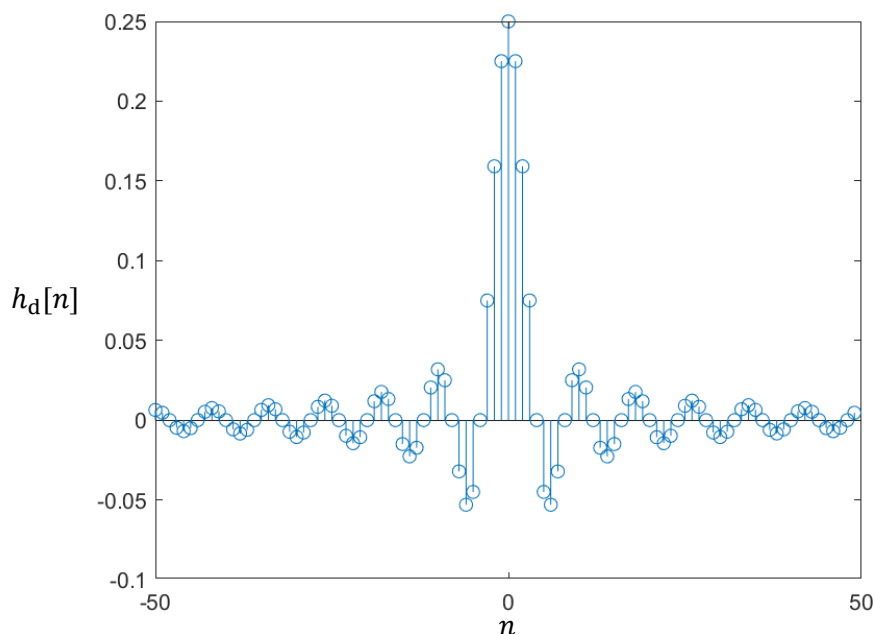
FIR Filter Design by Windowing

Suppose we want to design a low-pass filter with a cutoff frequency of ω_c . The desired frequency response will be:

$$H_d(\omega) = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & \text{otherwise} \end{cases}$$

To find the equivalent time-domain representation, we calculate the inverse discrete-time Fourier transform:

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{+\pi} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{+\omega_c} 1 \cdot e^{j\omega n} d\omega = \frac{\sin(n\omega_c)}{n\pi}$$



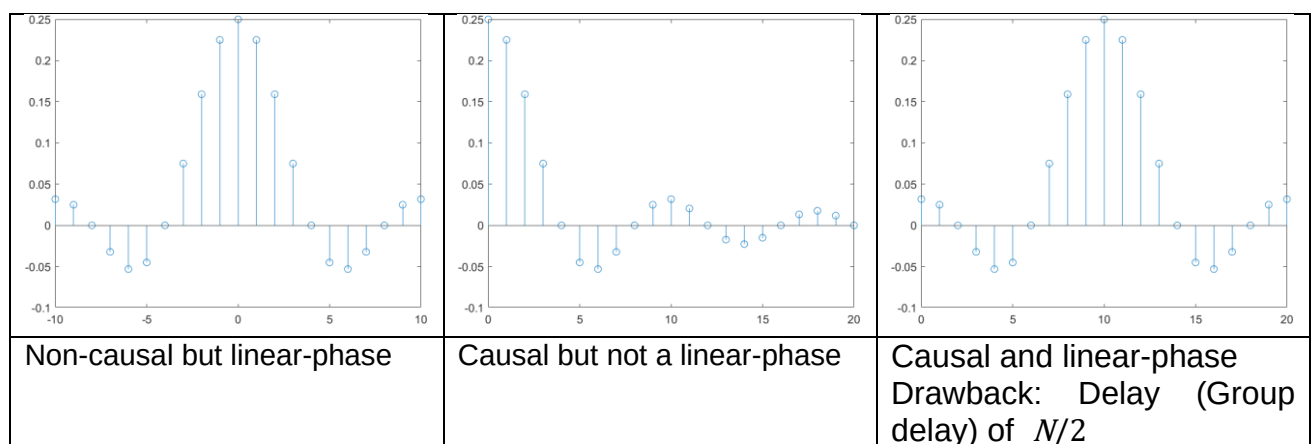
$$h_d[n] \text{ for } \omega_c = \frac{\pi}{4}.$$

It shows, that $h_d[n]$ is infinite in the time-domain and that the system is not a causal system.

The obvious solution will be to truncate the impulse response and use, for example, only 21 samples and assume the other coefficients to be zero. (→ Windowing with a Rectangular).

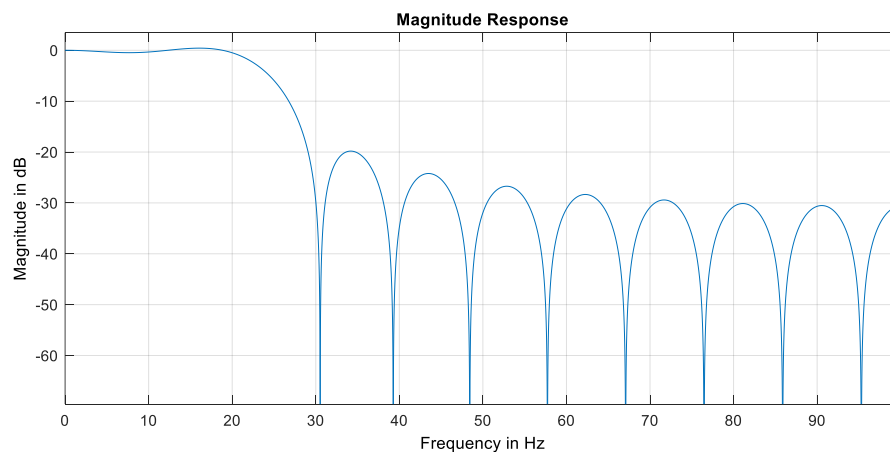
If we increase the number of samples, the truncated impulse response will be closer to the ideal impulse response and therefore the frequency response of the achieved filter will be close to the ideal filter. But as we increase the number of samples, more hardware will be required.

If we choose to use only 21 coefficients of the ideal impulse response, there will be three options which are shown in the following figures:



Truncation of the impulse response is equivalent to multiplying $h_d[n]$ (or its shifted version) by a rectangular window $w[n]$. Therefore, considering the applied shift, we obtain the impulse response of the designed filter: $h[n] = h_d\left[n - \frac{N}{2}\right] \cdot w[n]$.

Clearly, the spectrum of the rectangular window will cause the filter response to deviate from the ideal response.



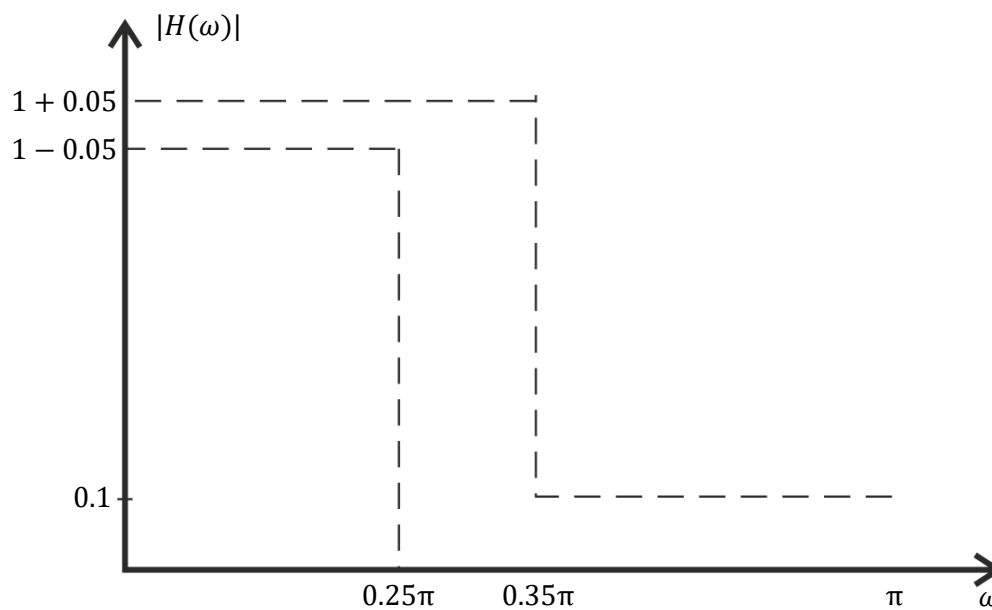
Question 1

An FIR low-pass filter is required to have the following specifications:

$$0.95 < |H(e^{j\omega})| < 1.05 \quad 0 \leq |\omega| < 0.25\pi$$

$$|H(e^{j\omega})| < 0.1 \quad 0.35\pi \leq |\omega| < \pi$$

Draw the tolerance scheme and determine which window functions are suitable to meet the above listed specifications. Assume that we would like to design the low-pass filter using the window method. Calculate the minimum order N of the low-pass filter for each suitable window function.



Filter requires a maximal passband error of $\delta_{\text{pass}} = 0.05$, and a maximal stopband error of $\delta_{\text{stop}} = 0.1$. Converting these values to dB results in

$$\delta_{\text{pass}} = 20 \cdot \log_{10}(0.05) = -26 \text{ dB},$$

$$\delta_{\text{stop}} = 20 \cdot \log_{10}(0.1) = -20 \text{ dB}.$$

This requires a window with a peak approximation error less than -26 dB.

→Hanning, Hamming, and Blackman windows meet this criterion.

The minimum order of the filter can be found using the approximate width of mainlobe, since the mainlobe width is about equal to the transition width, which is $0.35\pi - 0.25\pi = 0.1\pi$:

$$\text{Hanning: } 0.1\pi = \frac{8\pi}{N} \rightarrow N = 80, \text{ Hamming: } 0.1\pi = \frac{8\pi}{N} \rightarrow N = 80,$$

$$\text{Blackman: } 0.1\pi = \frac{12\pi}{N} \rightarrow N = 120.$$

Question 2

An FIR low-pass filter is required to have the following specifications:

- Pass-band edge frequency $f_{\text{pass}} = 2 \text{ kHz}$.
- Transition width $\Delta f = 200 \text{ Hz}$.
- Pass-band ripple $\delta_{\text{pass}} = 0.01$.
- Minimum stop-band attenuation $\delta_{\text{stop}} = 50 \text{ dB}$.
- Sampling frequency $f_s = 10 \text{ kHz}$.

We wish to design the digital filter using the window method. Determine an appropriate window function and calculate the required filter order to design the filter. Furthermore, derive the equation to determine the filter coefficients.

Convert the given specifications:

$$f_{\text{pass,normalized}} = \frac{f_{\text{pass}}}{f_s} = 0.2, \text{ or } \omega_{\text{pass,normalized}} = \frac{2\pi f_{\text{pass}}}{f_s} = 0.4\pi$$

$$\delta_{\text{pass}} = 20 \cdot \log_{10}(0.01) = -40 \text{ dB}$$

$$\Delta f_{\text{normalized}} = \frac{\Delta f}{f_s} = 0.02, \text{ or } \Delta \omega_{\text{normalized}} = \frac{2\pi \Delta f}{f_s} = 0.04\pi$$

Choose appropriate window:

Hamming, because it has the narrowest transition width, but still holds the δ_{stop} specification of -50 dB ($\delta_{\text{stop}} < \delta_{\text{pass}}$).

$$0.04\pi = \frac{8\pi}{N} \rightarrow N = 200$$

Derive the impulse response of the filter / equation to determine the filter coefficients:

$$h_d[n] = \frac{\sin(n\omega_c)}{n\pi}$$

$$\omega_c = \omega_{\text{pass,normalized}} + 0.5 \cdot \Delta \omega_{\text{normalized}} = 0.4\pi + 0.02\pi = 0.42\pi$$

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N}\right)$$

$$h[n] = h_d\left[n - \frac{N}{2}\right] \cdot w[n] = \frac{\sin\left(\left(n - \frac{N}{2}\right)\omega_c\right)}{\left(n - \frac{N}{2}\right)\pi} \cdot 0.54 - 0.46 \cos\left(\frac{2\pi n}{N}\right)$$

Question 3

By using the window method, a discrete-time filter is to be designed. The filter, which should meet the following specifications, is to be designed with a Kaiser window.

$$|H(e^{j\omega})| \leq 0.01 \quad 0 \leq |\omega| < 0.25\pi$$

$$0.95 \leq |H(e^{j\omega})| \leq 1.05 \quad 0.35\pi \leq |\omega| < 0.6\pi$$

$$|H(e^{j\omega})| \leq 0.01 \quad 0.65\pi \leq |\omega| < \pi$$

- Determine the minimum length $N + 1$ of the impulse response and the value of the Kaiser windows parameter β .
- Calculate the group delay.
- Determine the ideal impulse response $h_d[n]$ to which the Kaiser window should be applied to.

- Use the minimum specifications:

$$\delta = 0.01$$

$$\Delta\omega = 0.05\pi$$

$$A = -20 \log_{10}(\delta) = 40$$

$$N + 1 = \frac{A - 8}{2.285\Delta\omega} + 1 = 90.2 \rightarrow 91$$

$$\beta = 0.5842(A - 21)^{0.4} + 0.07886(A - 21) = 3.395$$

- Since it is a linear-phase filter with order 90, the group delay is $\frac{N}{2} = \frac{90}{2} = 45$ samples.
- There are two possible methods to calculate $h_d[n]$ of the band-pass filter, that are worth mentioning.

Inverse DTFT: $h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{+\pi} H_d(\omega) e^{j\omega n} d\omega$ $= \frac{1}{2\pi} \int_{-0.625\pi}^{-0.3\pi} 1 \cdot e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{+0.3\pi}^{+0.625\pi} 1 \cdot e^{j\omega n} d\omega$ $= \frac{\sin(0.625\pi n)}{\pi n} - \frac{\sin(0.3\pi n)}{\pi n}$	The impulse response of a band-pass filter can be found by subtracting the responses of two low-pass filters. $h_d[n] = \frac{\sin(0.625\pi n)}{\pi n} - \frac{\sin(0.3\pi n)}{\pi n}$
---	---

Question 4

The impulse response of an FIR filter is given as: $h[n] = [0.0565, 0.4435, 0.4435, 0.0565]$.

- Determine the difference equation of this filter system.
- Write down the expression of $H(e^{j\omega})$ and calculate the magnitude response $|H(e^{j\omega})|$ at $\omega = 0, 0.5\pi, \pi$.
- Is this filter a low-pass, band-pass or high-pass filter?
- The filter was designed using a rectangular window. Determine the maximum error in the passband and in the stopband. State the result linearly and in dB.
- Calculate the group delay of the filter.

a)

$$y[n] = 0.0565x[n] + 0.4435x[n-1] + 0.4435x[n-2] + 0.0565x[n-3]$$

b)

$$H(e^{j\omega}) = \sum_{n=0}^N h[n] \cdot e^{-j\omega n} = 0.0565 + 0.4435e^{-j\omega} + 0.4435e^{-2j\omega} + 0.0565e^{-3j\omega}$$

$$|H(\omega = 0\pi)| = |0.0565 + 0.4435 + 0.4435 + 0.0565| = 1$$

$$|H(\omega = 0.5\pi)| = |0.0565 + 0.4435(-j) + 0.4435(-1) + 0.0565(+j)| = 0.5473$$

$$|H(\omega = 1\pi)| = |0.0565 + 0.4435(-1) + 0.4435(+1) + 0.0565(-1)| = 0$$

c) Low-pass filter

d) Approximate error in the pass-band and in the stop-band are equivalent:

$$\delta = -21 \text{ dB}$$

$$\delta = 10^{-\frac{21}{20}} = 0.089$$

e) Group delay: $\frac{N}{2} = \frac{3}{2} = 1.5$ samples

FIR Filter Design – Cheat Sheet

Window Method

Window Type	Function	Peak Sidelobe Amplitude (Relative) in dB	Approximate Width of Main Lobe	Peak Approximation Error in dB
Rectangular	$w(n) = 1$	-13	$\frac{4\pi}{N+1}$	-21
Bartlett	$w(n) = 1 - \frac{2 n - \frac{N}{2} }{N}$	-25	$\frac{8\pi}{N}$	-25
Hann	$w(n) = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N}\right)$	-31	$\frac{8\pi}{N}$	-44
Hamming	$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N}\right)$	-41	$\frac{8\pi}{N}$	-53
Blackman	$w(n) = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N}\right) + 0.08 \cos\left(\frac{4\pi n}{N}\right)$	-57	$\frac{12\pi}{N}$	-74

Window Type	Function
Kaiser	$w(n) = \frac{I_0\left[\beta \sqrt{1 - \left(\frac{2n}{N-1}\right)^2}\right]}{I_0(\beta)}$ $I_0(x) = \sum_{k=0}^{\infty} \left(\frac{x^k}{2^k k!}\right)^2$ $A = -20 \log(\delta)$ $\Delta\omega = \omega_{\text{stop}} - \omega_{\text{pass}}$ $\beta = \begin{cases} 0.1102(A - 8.7) & A > 50 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & 21 \leq A \leq 50 \\ 0.0 & A < 21 \end{cases}$ $N = \frac{A - 8}{2.285\Delta\omega}$

Optimal Method

Minimum order N_e of the filter for the optimal method:

$$N_e = \frac{2}{3} \log\left(\frac{1}{10\delta_{\text{pass}}\delta_{\text{stop}}}\right) \cdot \frac{\omega_{\text{stop}}}{\omega_{\text{stop}} - \omega_{\text{pass}}}.$$

